

```
In [1]: import sys
print(sys.float_info)
```

```
sys.float_info(max=1.7976931348623157e+308, max_exp=1024, max_10_exp=308, min
=2.2250738585072014e-308, min_exp=-1021, min_10_exp=-307, dig=15, mant_dig=5
3, epsilon=2.220446049250313e-16, radix=2, rounds=1)
```

```
In [2]: import numpy as np
print(np.finfo(np.float))
```

Machine parameters for float64

```
-----
precision = 15    resolution = 1.0000000000000001e-15
machep = -52     eps = 2.2204460492503131e-16
negep = -53     epsneg = 1.1102230246251565e-16
minexp = -1022   tiny = 2.2250738585072014e-308
maxexp = 1024    max = 1.7976931348623157e+308
nexp = 11       min = -max
-----
```

```
In [4]: print(np.iinfo(np.int))
```

Machine parameters for int64

```
-----
min = -9223372036854775808
max = 9223372036854775807
-----
```

```
In [4]: np.finfo(np.float64).__dir__()
```

```
Out[4]: ['dtype',
         'precision',
         'iexp',
         'maxexp',
         'minexp',
         'negexp',
         'machep',
         'tiny',
         'resolution',
         'epsneg',
         'bits',
         'max',
         'min',
         'eps',
         'nexp',
         'nmant',
         'machar',
         '_str_tiny',
         '_str_max',
         '_str_epsneg',
         '_str_eps',
         '_str_resolution',
         '__module__',
         '__doc__',
         '_finfo_cache',
         '__new__',
         '_init',
         '__str__',
         '__repr__',
         '__dict__',
         '__weakref__',
         '__hash__',
         '__getattr__',
         '__setattr__',
         '__delattr__',
         '__lt__',
         '__le__',
         '__eq__',
         '__ne__',
         '__gt__',
         '__ge__',
         '__init__',
         '__reduce_ex__',
         '__reduce__',
         '__subclasshook__',
         '__init_subclass__',
         '__format__',
         '__sizeof__',
         '__dir__',
         '__class__']
```

```
In [5]: np.finfo(np.float64).bits
```

```
Out[5]: 64
```

```
In [5]: import numpy as np
print(np.finfo(np.float32))
```

Machine parameters for float32

```
-----
precision =    6    resolution = 1.0000000e-06
machep  =   -23    eps =      1.1920929e-07
negexp  =   -24    epsneg =    5.9604645e-08
minexp  =  -126    tiny =      1.1754944e-38
maxexp  =    128    max =      3.4028235e+38
nexp    =     8    min =      -max
-----
```

```
In [6]: print(np.iinfo(np.int32))
```

Machine parameters for int32

```
-----
min = -2147483648
max = 2147483647
-----
```

```
In [ ]:
```